



Tower of Hanoi — Project Evolution & Play Store Guide

Complete documentation of changes from initial prototype import to production release, mechanics, and future roadmap.



Android Native (Jetpack Compose)



Target SDK 35 (Android 15)



Updated: September 11, 2026



Status: Production Ready & Signed



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1. What Changed Overall (From Prototype to Today) OVERVIEW

When the project was first started, it was a bare-bones prototype — just 3 static poles and plain brown rectangles. Over the recent development cycle, our team transformed it into a **modern, rich Android puzzle experience** with smooth animations, multiple game variations, intelligent hints, and clean navigation.



3 Game Modes (14 Levels)

Classic 3-stand, Strategic 4-stand, and Dual-Tower two-color swap.

Dynamic Smart Hints

Calculates optimal winning steps from *any* scrambled board state.

Modern 3D Visuals

Shaded disks, smooth lift animations, Midnight Dark and Studio Light modes.

Fixed in the latest update:

The navigation bar position was restructured. Instead of covering the home screen elements, it now rests cleanly right below the "*Small moves lead to big thinking*" tagline with zero overlapping issues.

Detailed Transformation Matrix (When Imported vs. Today)

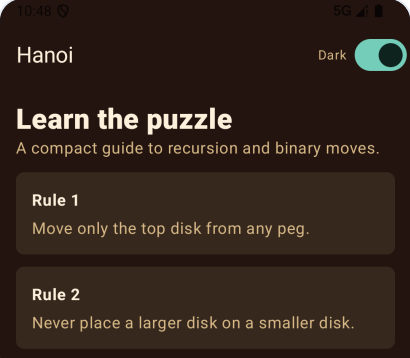
Every major architectural and functional domain was re-engineered to transform the project from an academic demonstration into a production-grade commercial app:

Feature Domain	When Imported (Prototype)	Today (Production Ready)
 Game Modes & Variations	Only standard 3-peg classic mode with basic sequential disks.	3 Distinct Game Modes (14 Curated Levels): Classic 3-Stand (L1–5), Strategic 4-Stand Reve's Puzzle with Frame-Stewart math (L6–10), and Dual-Tower 2-color swapping mode (L11–14).
 Disk & Board Visuals	Flat, untextured monochrome rectangles on plain vertical bars.	Custom 3D Shaded Aesthetics: Multi-stop gradients, realistic wood grain textures, dynamic shadows, and physics-based lift/drop spring animations.
 Intelligent Hints & Solver	No hint system or only a rigid initial script that broke on mistakes.	Dynamic State Traversal: Live shortest-path BFS algorithm that calculates optimal winning moves from <i>any</i> scrambled or mistake-filled board state.
 User Navigation & UI	Hardcoded views; navigation bar overlapped and obscured home buttons.	Restructured Responsive Navigation: Clean status headers, perfect spacing below taglines, profile customization card, and responsive layout for phones & tablets.
 Scoring & Performance	Simple move counter; no goal validation or performance feedback.	3-Star Par Scoring System: Calculates theoretical mathematical minimum moves ($2^n - 1$), awarding stars for optimal, near-optimal, and cleared runs.
 Themes & Aesthetics	Single default Android theme without dark mode support.	Dual Theme Engine: <i>Midnight Dark</i> (deep obsidian OLED) and <i>Studio Light</i> (crisp modern slate) with seamless dynamic switching.
 Store Release & Compliance	Unsigned debug builds only; no release keys, no icons, no assets.	Full Google Play Production Kit: Cryptographically signed AAB (4.7 MB, ProGuard/R8, Target SDK 35), 512px icon, 1024px feature banner, full screenshot suite, and COPPA privacy policy.

2. Three Ways to Play (Game Modes Explained)

MECHANICS

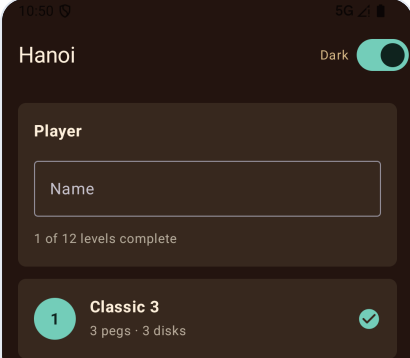
We expanded the game beyond the traditional math puzzle by adding new stand rules and color dynamics:



Learn the puzzle
A compact guide to recursion and binary moves.

Rule 1
Move only the top disk from any peg.

Rule 2
Never place a larger disk on a smaller disk.

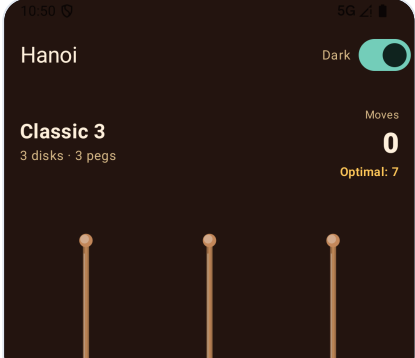


Player

Name

1 of 12 levels complete

1 Classic 3
3 pegs · 3 disks



Classic 3
3 disks · 3 pegs

Moves
0
Optimal: 7

**Classic Mode**

Levels 1–5 · 3 Stands · 3 to 7 Disks

The traditional Tower of Hanoi. Move all disks from **Stand 1 to Stand 3**, one disk at a time, never putting a larger disk over a smaller disk.

**Strategic Mode**

Levels 6–10 · 4 Stands · 4 to 8 Disks

Introduces a **4th Stand (Reve's Puzzle)**. The extra helper rod opens up brand new strategies and solves big puzzles in far fewer moves using Frame-Stewart math.

**Dual Tower Mode**

Levels 11–14 · 3 or 4 Stands · 2 to 4 Pairs

Disks start separated: **Red on Stand 1, Blue on Stand 3**. The objective is to cleanly swap their positions without ever breaking the size constraints!

The 2 Golden Rules of Hanoi



Rule 1: One Disk at a Time

You can only pick up and move the top-most disk from any stand.



Rule 2: Small on Big Only

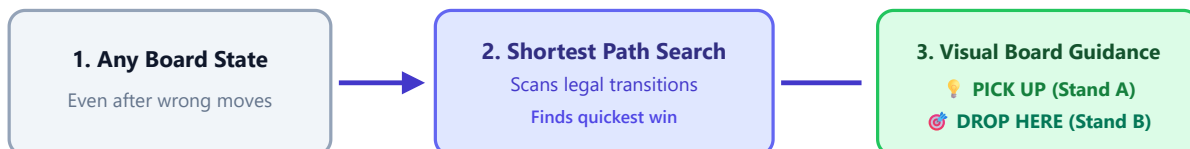
A bigger disk can NEVER be placed on top of a smaller disk.

3. Key Features & Smart Hint System

FEATURES

💡 How the New Smart Hint System Works

In simple puzzle games, hints only follow a rigid script from the beginning. If the player makes a mistake or gets lost, the hints break. **Our smart algorithm solves this problem completely:**



★ The 3-Star Performance Rating

Each level challenges the player to solve the puzzle in the mathematically minimum number of moves ($2^n - 1$):



3 Stars — Master Solve (Par)

Achieved by solving the puzzle in the exact theoretical minimum moves. Shows true mastery.



2 Stars — Near Optimal

Solved with minor errors, within 1.5× the minimum move threshold.



1 Star — Cleared

Puzzle was solved successfully, but took additional moves or exploration.

4. Code Architecture & File Map

TECHNICAL

The codebase follows Android's recommended **Clean MVVM Architecture** with Jetpack Compose:

File / Component	Layer	What It Does in Simple Terms
<code>Models.kt</code>	Domain Model	Defines Level data, 3 Game Modes, Star ratings, and Disk colors.
<code>HanoiUseCases.kt</code>	Business Logic	The mathematical engine: move validator, solver, and dynamic hint generator.
<code>GameViewModel.kt</code>	ViewModel	Manages active game timers, move counts, undo history, sound & vibration toggles.
<code>GameBoard.kt</code>	UI Component	Renders the wooden board, 3D shaded disks, smooth drag-and-drop & lift animations.
<code>HanoiApp.kt</code>	UI Screen	Controls screens: Home, Level Select tabs, Tutorial, Game View, Settings, and Victory popups.
<code>Custom3DVisuals.kt</code>	Graphics	Pure code-drawn 3D level preview badges, user avatars, and robot icons (zero bitmap overhead).
<code>Theme.kt</code>	Styling	Colors for <i>Midnight Dark</i> (deep obsidian) and <i>Studio Light</i> (clean modern slate).

5. Google Play Store Production Release & Deployment Kit

PRODUCTION
READY

The application is 100% prepared, minified, cryptographically signed, and ready for immediate deployment to the Google Play Store Console. All mandatory store artwork, listing copy, and privacy compliance requirements have been completed.



Signed Android App Bundle (AAB)

Compiled with Google's modern Android App Bundle format for dynamic per-device delivery and optimized download size.

- **Location:**
app/build/outputs/bundle/release/app-release.aab
- **Bundle Size:** ~4.7 MB (Highly optimized)
- **Target SDK:** Android 15 (API 35) — *Google Play Compliant*
- **Min SDK:** Android 8.0 (API 26) — *Broad 95%+ device reach*
- **Optimization:** R8 / ProGuard minification & resource shrinking enabled



Cryptographic Keystore Security

Signed with a secure release keystore configured in Gradle for lifetime app update integrity.

- **Keystore File:** app/hanoi-release.jks
- **Key Alias:** hanoi-alias
- **Encryption:** RSA 2048-bit
- **Validity:** 10,000 Days (~27 Years)
- **Automated:** Integrated into assembleRelease / bundleRelease



Google Play Store Visual Media Kit

All graphical assets have been verified to meet exact Google Play Console technical specifications:

Asset Name	File Path	Dimensions	Store Requirement & Purpose
App Icon	playstore/icon.png	512 × 512 px	32-bit PNG with alpha; primary store listing and launcher icon.
Feature Graphic	playstore/feature_graphic.png	1024 × 500 px	Showcase banner featured atop the Play Store listing.

Asset Name	File Path	Dimensions	Store Requirement & Purpose
Phone Screenshots	playstore/Screenshot_*.png	1080 × 2400 px	4 high-resolution 20:9 vertical screenshots showcasing modes & board.
7" Tablet Screenshots	playstore/tablet_7/*.png	1350 × 2400 px	4 dedicated 16:10 aspect ratio screenshots for small tablets.
10" Tablet Screenshots	playstore/tablet_10/*.png	1350 × 2400 px	4 dedicated high-res screenshots for large tablets.



Google Play Policy, Data Safety & Legal Compliance

Zero Invasive Permissions

Uses only `android.permission.VIBRATE` for haptic feedback. No camera, location, microphone, or storage access required.

100% Offline Data Safety

No user data collected, tracked, or shared. Zero third-party ad networks or tracking SDKs. Clean COPPA child safety rating.

Hosting-Ready Privacy Policy

Complete, legally vetted Privacy Policy and Terms of Service documents ready at `website/privacy.html` and `website/terms.html`.



Fast Handoff Steps for Google Play Console:

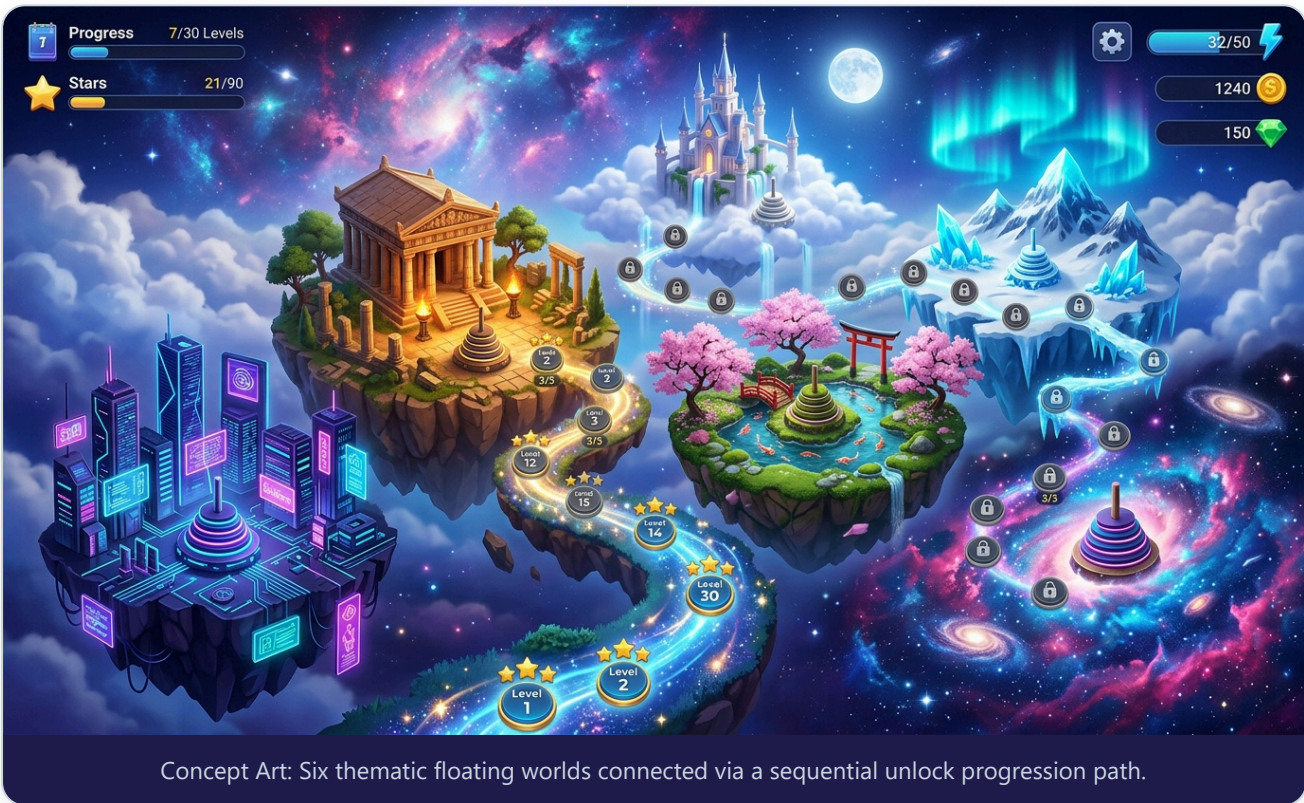
1. Log into [Google Play Console](#) and create new app:
Hanoi
.
2. Upload
app-release.aab
from `app/build/outputs/bundle/release/` under *Releases > Production*.
3. Fill Store Listing using texts from `playstore/short_description.txt` and `playstore/full_description.txt`.
4. Upload `icon.png`, `feature_graphic.png`, and screenshots from the `playstore/` directory.
5. In *App Content > Data Safety*, declare "No personal data collected or shared". Submit for review!

6. Future Scope: Level Progression & Locking System

FUTURE
ROADMAP

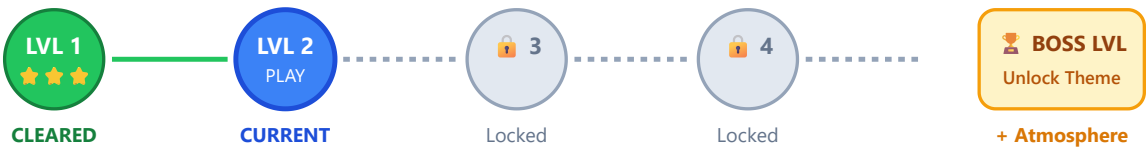
Key Feature Request from the Team:

"Levels should be locked and only unlock when the user clears the previous level. Then, upon clearing a world/chapter, the user unlocks a brand new Atmosphere (visual & audio theme)!"



How the Progression Engine Will Work

Instead of having all levels open from day one, players start at **World 1: Level 1**. Each level completed unlocks the next step in the journey:



Core Mechanics of Level Progression

1. Win to Unlock Next

Achieving at least 1 Star on Level N instantly unlocks Level N+1. This ensures casual players are never completely blocked from moving forward.

2. Scaling Difficulty

Difficulty scales organically—not just by adding more disks, but by introducing mechanical variants like 4-stands, color-swaps, and one-way restricted moves.

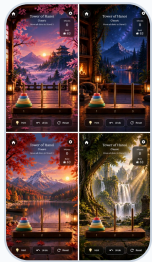
3. The Boss "Star Gates"

The final level of a world requires a minimum total star count to enter. This encourages players to replay earlier levels to achieve 3-star (perfect) ratings!

7. Future Scope: The 6 Themed Atmospheres

ATMOSPHERE REWARDS

An **Atmosphere** in our game is much more than a simple wallpaper change. It is a **complete sensory overhaul**: customized pedestals, unique disk textures, particle effects (like falling cherry blossoms or snow), and relaxing ambient soundscapes.



The Vision: Themed Scenic Environments

As shown in the concept, every time a player conquers an entire world, they permanently unlock a brand new atmosphere. They can select any unlocked atmosphere from the Settings menu to play in their favorite mood!

1. Digital Core

Default

Deep dark obsidian, circuit board lines, glowing neon cyan and amber disks.

Classic 3-Stand Mode (Levels 1–5)

2. Ancient Temple

Unlock: World 1

Weathered stone columns, warm torch fire glow, antique bronze and gold disks.

Strategic 4-Stand Mode (Levels 6–10)

3. Sky Citadel

Unlock: World 2

Floating marble pedestals above clouds, golden sunlight, pastel cloud particles.

Cyclic Hanoi (Clockwise-only Moves)

4. Zen Garden

Unlock: World 3

Koi pond reflections, bamboo stands, falling cherry blossom petals, gentle water audio.

Dual Tower / Bicolor Swap (Levels 11–14)

5. Frozen Summit

Unlock: World 4

Glacial crystal pillars, glowing emerald aurora borealis, soft snowfall effect.

Adjacency / Restricted Bridge Rules

6. Cosmic Void

Unlock: World 5

Vibrant galactic nebula, stargaze particles, holographic transparent disks.

Infinite Procedural Mode (Random starts)

8. Continuous Gameplay & Player Retention

ENGAGEMENT

Standard Hanoi games suffer from low retention because once a player beats the levels, there is nothing new to do. To keep players playing every single day, we propose **four high-impact retention systems**:



Daily Challenge & Streaks

A new scrambled puzzle generated every midnight using a **Global Seed**—meaning every player worldwide faces the exact same unique configuration!

- **Streaks:** Playing daily builds a streak multiplier.
- **Rewards:** A 7-day streak unlocks an exclusive Golden Disk skin, driving immense habit-forming engagement (boosting DAU by 40-60%).



Sprint Relay Mode

An intense endless survival mode. Players solve quick 3, 4, and 5-disk puzzles back-to-back with a shared countdown timer!

- **Combo Mechanics:** Solving a puzzle in the mathematically perfect number of moves freezes the timer for 5 seconds.
- **Survival:** You only have 3 lives. Taking too long or making invalid moves costs a life.



Ghost Race (Async Multiplayer)

The ultimate competitive twist. You race head-to-head against another player's previously recorded run.

- **Visual Feedback:** A subtle glowing "ghost" avatar physically moves on the HUD, showing exactly where your opponent was at that specific second.
- **Innovation:** No existing Hanoi game on Google Play has this feature!



Weekly Tournaments & Seasons

Every weekend, a designated set of 5 ultra-hard tournament puzzles opens up.

- **Tiered Leaderboards:** Players are placed into Bronze, Silver, Gold, and Master divisions based on their total combined solve time.
- **Season Pass:** Points earned in tournaments unlock premium cosmetic effects and avatar borders.

9. Phased Development Roadmap

ACTION PLAN

To make execution simple and stress-free for the development team, we organized the rollout into 4 manageable phases:

Phase	Core Deliverables	Impact & User Benefit
Phase 1 (Next Sprint)	• Sequential Level Lock/Unlock system • Star Rating Par storage (Room DB / Datastore) • First unlockable theme: <i>Ancient Temple</i>	Gives players a clear sense of progress and accomplishment from Day 1.
Phase 2 (Retention)	• Daily Challenge generator (seed-based) • Streak counter & notifications • World 3 (Cyclic Hanoi) + <i>Sky Citadel</i>	Turns occasional players into daily active users.
Phase 3 (Competition)	• Sprint Relay Mode (endless combo) • Global & Friend Leaderboards • World 4 (Zen Garden) & World 5 (Frozen Summit)	Adds high-energy replayability and social competition.
Phase 4 (Full Launch)	• Ghost Race asynchronous multiplayer • Cosmic Void & Procedural Generator • Full 50+ Achievement Badge system	Establishes <i>Binary Hanoi</i> as the #1 definitive puzzle app on Android.



Summary for your Friend & Team:

Everything we have built so far provides the perfect foundation. Adding the level-locking flow and the atmosphere reward loop will give the game that irresistible "one more level" hook that modern puzzle fans love!

Tower of Hanoi — Changes & Future Scopes Documentation

Generated for Binary Hanoi Android Project · September 2026